

Hefang Deng

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RESEARCH INTERESTS

My research interest lies in turbomachinery aerodynamics, next-generation advanced propulsion, and AI for Science. Currently, my research focuses on leveraging AI to advance and elevate aerodynamic design capabilities for advanced turbomachinery.

EDUCATION

Shanghai Jiao Tong University

Doctor of Philosophy in Aerospace Engineering

- **Advisor:** Prof. Jinfang Teng

Shanghai, China

Apr 2021 – Dec 2024

Harbin Engineering University

Master of Engineering in Power Engineering and Engineering Thermophysics

- **Advisor:** Prof. Yuting Jiang

Harbin, China

Sep 2018 – Apr 2021

Harbin Engineering University

Bachelor of Engineering in Energy and Power Engineering

- **Advisor:** Prof. Qun Zheng

Harbin, China

Sep 2014 – Jul 2018

RESEARCH PROJECTS

Shanghai Municipal Education Commission Joint Innovation Program

Research on Tip Leakage Flow Mechanism in High-Pressure Compressors

Apr 2021 – Dec 2024

National Natural Science Foundation of China (General Program)

Evolution and Control Mechanism of Clearance Flow Field in Cantilevered Stator with High-Speed Rotating Hub

Apr 2021 – Dec 2024

National Science and Technology Major Project (Aero-Engine & Gas Turbine)

Theoretical and Methodological Research on Aerodynamic Stability Analysis of Multi-Stage Compressors

Apr 2021 – Dec 2024

National Natural Science Foundation of China (Major Program)

Research on Fault Precursor Enhancement and Fusion Early Warning Method for Large Civil Aeroengine Ground Test

Jan 2024 – Dec 2024

SELECTED PUBLICATIONS

- [1] **Hefang Deng**, Songan Zhang, Honglin He, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. Design optimization methods for a compressor rotor under inlet distortion. *Aerospace Science and Technology*, Vol. 167, 109987, 2026.
- [2] **Hefang Deng**, Huihui Li, Songan Zhang, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. Aerodynamic performance in a boundary layer ingesting fan with swirl distortion. *Aerospace Science and Technology*, Vol. 148, 109102, 2024.
- [3] Honglin He, **Hefang Deng**, Xiang Zuo, Songan Zhang, Jinfang Teng, Mingmin Zhu. Comparative study of neural structure paradigms for compressor cascade flow field modeling based on the condition-encoder. *Aerospace Science and Technology*, Vol. 166, 109898, 2026.
- [4] Kailong Xia, **Hefang Deng**, Mingmin Zhu, Huayin Chen, Jinfang Teng. A novel dynamic mesh method for multi-scenario turbomachinery simulations: Development, validation, and application. *Aerospace Science and Technology*, Vol. 169, 110120, 2026.
- [5] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. Partial rotating instability in a boundary layer ingesting fan. *Chinese Journal of Aeronautics*, Vol. 38, No. 3, 100217, 2025.
- [6] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. On the mechanism of rotating stall in a boundary layer ingesting fan. *Journal of Engineering for Gas Turbines and Power*, Vol. 147, No. 9, 091018, 2025.

- [7] Yaqi Liu, **Hefang Deng**, Jinfang Teng, Mingmin Zhu, Songan Zhang, Shaopeng Lu. Loss mechanisms of leakage flow in a cantilevered stator with/without moving end wall. *Physics of Fluids*, Vol. 37, No. 3, 035131, 2025.
- [8] **Hefang Deng**, Songan Zhang, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng. Performance prediction and design optimization of a transonic rotor based on deep transfer learning. *Physics of Fluids*, Vol. 36, No. 9, 097107, 2024.
- [9] Kailong Xia, **Hefang Deng**, Yongjian Zhong, Mingmin Zhu, Xiaoqing Qiang, Jinfang Teng. The impact of second-harmonic tip clearance distributions caused by casing ovalization and blade height variation in a large-scale compressor stage. *Aerospace Science and Technology*, Vol. 149, 109180, 2024.
- [10] Xiaoqing Qiang, **Hefang Deng**, Kailong Xia, Jinfang Teng, Mingmin Zhu, Shaopeng Lu. Effects of moving endwall on the unsteadiness of tip leakage flow in compressor cascades. *Physics of Fluids*, Vol. 35, No. 10, 105121, 2023.
- [11] **Hefang Deng**, Kailong Xia, Xiaoqing Qiang, Mingmin Zhu, Jinfang Teng, Huihui Li. Investigation on the inflow distortion of a propulsive fuselage aircraft and its impact on the propulsion fan performance. *Physics of Fluids*, Vol. 35, No. 8, 085122, 2023.
- [12] **Hefang Deng**, Kailong Xia, Jinfang Teng, Xiaoqing Qiang, Mingmin Zhu, Shaopeng Lu. An experimental facility with the high-speed moving endwall for axial compressor leakage flow research. *Aerospace*, Vol. 10, No. 5, 432, 2023.
- [13] **Hefang Deng**, Kailong Xia, Jinfang Teng, Shaopeng Lu, Mingmin Zhu, Xiaoqing Qiang. Loss analysis of cavity leakage flow in a compressor cascade. *Journal of Turbomachinery*, Vol. 144, No. 12, 121008, 2022.
- [14] **Hefang Deng**, Jinfang Teng, Mingmin Zhu, Xiaoqing Qiang, Shaopeng Lu, Yuting Jiang. Overall cooling performance evaluation for film cooling with different winglet pairs vortex generators. *Applied Thermal Engineering*, Vol. 212, 118560, 2022.

HONORS & AWARDS

Shanghai Super Postdoctoral Incentive Program	2025
China Postdoctoral Science Foundation (Grade C)	2025
GPPS Excellent Paper Award (4/114)	2024
Outstanding Participant Award, Aerospace Doctoral Competition	2024

ACADEMIC SERVICE

Journal Reviewer:

- *Aerospace Science and Technology*
- *Physics of Fluids*
- *GPPS Journal*
- *Nuclear Engineering and Design*
- *International Journal of Turbo & Jet-Engines*
- *Journal of Engineering for Thermal Energy and Power*

Academic Appointments:

- Session Chair, Compressor Session, GPPS Conference, 2023
- Early-Career Editorial Board Member, *Journal of Engineering for Thermal Energy and Power*